NextGeneration Class

Deployment guide

Buy the server and the domain, then put the platform online, step by step.

≈ $15	$5.35	100	≈ 1 h
first month (server + domain)	per month after, no commitment	users at most, 4 GB is enough	of your time, plus OVH waiting

Part A	Before you start: what you need
Part B	Buy: SSH key, OVHcloud account, VPS-1 server, domain (steps 1 to 5)
Part C	Put it online: connect, domain, deploy, first sign-in, alerts (steps 6 to 11)
Part D	Living with it: updates, backups, restore, problems, end of the test
Part E	Safety rules

Decided on 12 November 2026 · Host: OVHcloud VPS-1, France, monthly · Web platform only · Strategy and technical details: `docs/04-delivery/05-DEPLOYMENT.md`

Prices are before VAT and may change slightly: always check the cart before paying.

PART A

Before you start

What you need

- [ ] A **Visa or Mastercard that pays in dollars or euros** (not a prepaid card if possible).
- [ ] Your **ID card or passport**: OVHcloud may ask for it once to verify a new account.
- [ ] An **email address** and a **phone number** you use every day (OVH invoices and down-site alerts go there).
- [ ] This **Mac**, with the project folder `NextGenerationClass` and Flutter installed (already the case).
- [ ] A **name for the website**, for example `ngclass.com`. Short, easy to say, no hyphen. Have 2 or 3 ideas in case the first is taken.
- [ ] A **password manager** or a safe notebook, for the owner password you will choose.

What you will buy

What	Choice	Price (before VAT)
Server	OVHcloud **VPS-1**: 2 vCores, 4 GB RAM, 40 GB NVMe, daily backup included, unlimited traffic, France	about **$5.35 a month**, monthly, cancel anytime
Domain	a **.com** at OVHcloud	about **$9.20** the first year, then about $15 to $16.50 a year
HTTPS, backups, alerts	Let's Encrypt, our nightly backup, UptimeRobot	**free**

> **How it works in one sentence**
>
> You buy the server and the domain, link the domain to the server, then one command on this Mac installs and starts everything: the platform, HTTPS, the firewall, the nightly backups.

> **Never share these, with anyone, including in a chat**
>
> Your OVHcloud password, the owner password of the platform, and the file `~/.ssh/ngclass_ovh` (the secret key). Only the server's **IP address** and the **domain name** can be shared.

PART B · BUY

Buy the server and the domain

1 Make a key to enter the server (2 min)

A key is safer than a password. Open the **Terminal** app on the Mac and run:

```
ssh-keygen -t ed25519 -f ~/.ssh/ngclass_ovh -C "ngclass-ovh"
```

It asks for a passphrase: choose one you will remember, or press Enter twice for none. Then copy the **public** part (safe to paste on OVH):

```
pbcopy < ~/.ssh/ngclass_ovh.pub
```

It is now in your clipboard, ready for step 3. The other file, `ngclass_ovh` without `.pub`, stays secret on this Mac.

2 Create your OVHcloud account (5 min)

1. Go to **ovhcloud.com/fr** and click **Créer un compte** (or start an order: it asks you to create one).
2. Account type: **Particulier**. No company is needed.
3. Name and address **exactly as on your ID**, country **Tunisie**, your real email and phone.
4. Turn on **two-step login** later in the account security settings (recommended).

3 Order the VPS-1 server (5 min)

Go to **ovhcloud.com/fr/vps**, find **VPS-1** and click **Commander**. Choose:

Screen	Choose
Modèle	**VPS-1** (2 vCores, 4 Go, 40 Go)
Engagement	**Sans engagement** (monthly). Not 12 months.
Localisation	**Europe** › **France** (Gravelines, Strasbourg or Roubaix)
Image / système	**Distribution seule** › **Ubuntu 24.04**. No apps, no panel (no cPanel, no Plesk).
Clé SSH	Paste the key from step 1 (Cmd + V)
Options	**Add nothing.** The daily backup is already included. No extra storage, no Windows.

Check the total is about **$5.35 a month** before VAT, then pay.

> **No "Clé SSH" field on your screen?**
> No problem: OVH will email a password instead. At step 6 you will use it once to install your key.

4 Order the domain (3 min)

1. Go to **ovhcloud.com/fr/domains** and search your name, for example `ngclass.com`.
2. Add the **.com** to the cart. **Add no options**: no email, no hosting, no paid "DNS anycast".
3. Keep **DNS OVHcloud** as it is, and pay with the same account.

You can put the server and the domain in the same cart.

5 Wait for the emails (a few minutes to a day)

- If OVH asks for an **ID check**, upload your ID card or passport. It usually takes a few hours.
- You will receive an email like **Votre VPS est prêt**. Note the **IPv4 address** (like `51.xx.xx.xx`) and the user name, usually `ubuntu`.
- The domain is usually active within an hour.

PART C · PUT IT ONLINE

Configure, step by step

In every command below, replace `IP` by your server address and `ngclass.com` by your domain.

6 Check you can enter the server (2 min)

```
ssh -i ~/.ssh/ngclass_ovh ubuntu@IP
```

The first time, type `yes`. If you see a line like `ubuntu@vps-…:~$`, it works. Type `exit` to leave.

OVH sent a password instead of using your key?

Run this once and type the emailed password when asked. Then the command above works without a password.

```
ssh-copy-id -i ~/.ssh/ngclass_ovh.pub ubuntu@IP
```

If OVH asks you to choose a new password at the first login, choose one and keep it in your password manager.

7 Point the domain to the server (5 min, then wait 5 to 30 min)

1. In the OVHcloud panel: **Web Cloud › Noms de domaine › ngclass.com › Zone DNS**.
2. Find the line of type **A** with an empty sub-domain. Click **…** › **Modifier l'entrée** and set the target to **your server IP**.
3. Do the same for the line **A** with sub-domain **www**. If there is none, **Ajouter une entrée** › **A**, sub-domain `www`, target your IP.
4. Delete any line of type **AAAA** (they point to an OVH parking page and would block the HTTPS certificate).

After 5 to 30 minutes, check on the Mac. Both must show your server IP:

```
dig +short ngclass.com
dig +short www.ngclass.com
```

Do not go to step 8 before both lines show your IP

The HTTPS certificate is requested at the first start. If the domain does not point to the server yet, it fails and retries later, which is slower.

8 Install and start the platform (one command, about 15 min)

In the Terminal on the Mac, go to the project folder and run the deploy command:

```
cd ~/Desktop/NextGenerationClass
./infra/deploy.sh ubuntu@IP
```

What happens, in order (you only answer the questions):

Step on screen	What it does	Time
1/5 Building the web app	Builds the website on the Mac	about 1 min
2/5 and 3/5	Creates `/opt/ngclass` on the server and sends the project	under 1 min
4/5 first time only	Updates Ubuntu, installs Docker, firewall, protection against password guessing, 2 GB swap, automatic security updates, nightly backup at 02:00	3 to 5 min
4/5 questions	Asks the **domain**, the **owner username**, **first name**, **last name** and a **first password** (typed twice, hidden). The database password and the security secret are generated on the server.	1 min
5/5 Starting	Builds and starts the database, the API and the web server, gets the HTTPS certificate	about 5 min

It ends with `Done: https://ngclass.com is up.`

> **The owner account**
> The owner (you) uses the admin dashboard to add teachers. The first password is temporary: the platform asks you to choose your own at the first sign-in. Write the username somewhere safe.

9 First sign-in (5 min)

1. Open **https://ngclass.com**. The lock icon in the address bar shows HTTPS works.
2. Sign in with the owner username and the first password, then choose your own password.
3. In the dashboard, **add the teacher**. The platform shows her username and a temporary password: give them to her privately. She chooses her own password at her first sign-in.
4. Check on a phone too: open the same address in the browser. In Chrome or Safari, **Add to Home Screen** gives an app-like icon.

10 Turn on the down-site alert (5 min, free)

1. Create a free account at **uptimerobot.com**.
2. **New monitor** › type **HTTP(s)** (or **Keyword** with the word `UP`).
3. URL: `https://ngclass.com/health`, interval **5 minutes**, alert to your email.

The `/health` address answers `UP` only when the website, the API and the database all work. If something stops, you get an email within about 5 minutes.

11 Check the OVH daily backup (1 min)

In the panel: **Bare Metal Cloud › Serveurs privés virtuels › your VPS**. In the backup section (**Sauvegarde automatisée**), check it is **active**. It is included in VPS-1: if it shows inactive and free, activate it. Together with our own nightly backup, the data is protected twice.

> **Done: the platform is online**
> HTTPS, the firewall, the automatic restarts, the nightly backups and the down-site alert are all in place.

PART D

Living with it

Put a new version online

The same command as step 8. It skips the server setup and the questions, rebuilds and restarts. Users see the new version at their next visit; the outage is under a minute.

```
cd ~/Desktop/NextGenerationClass
./infra/deploy.sh ubuntu@IP
```

Best done in the evening, when nobody is using it.

Look at the server

```
# enter the server
ssh -i ~/.ssh/ngclass_ovh ubuntu@IP
cd /opt/ngclass/infra

# state of the three parts (all should say "Up")
sudo docker compose -f docker-compose.prod.yml ps

# last lines of the API log
sudo docker compose -f docker-compose.prod.yml logs --tail 100 api

# restart everything
sudo docker compose -f docker-compose.prod.yml restart

# free disk and memory
df -h /
free -h
```

Backups

Every night at 02:00 (Tunis time) the server saves the database and all uploaded files in `/var/backups/ngclass`, and keeps 30 days.

```
# list the backups and read the backup log (on the server)
ls -lh /var/backups/ngclass
tail /var/log/ngclass-backup.log

# copy the latest backups to the Mac (run on the Mac), once a month for example
mkdir -p ~/ngclass-backups
scp -i ~/.ssh/ngclass_ovh "ubuntu@IP:/var/backups/ngclass/*" ~/ngclass-backups/
```

Restore a backup

Only if data was lost or deleted by mistake. Replace `DATE` by the backup to restore (see `ls /var/backups/ngclass`). On the server, in `/opt/ngclass/infra`:

```
C="sudo docker compose -f docker-compose.prod.yml"
$C stop api
$C exec -T db pg_restore --clean --if-exists -U ngclass -d ngclass < /var/backups/ngcl
ass/db_DATE.dump
$C run --rm --no-deps --user root -v /var/backups/ngclass:/backup --entrypoint sh api
\
  -c "rm -rf /data/files/* && tar -xzf /backup/files_DATE.tar.gz -C /data"
$C start api
```

If the whole server is lost: OVH can restore its daily backup from the panel, or order a new VPS-1 and run step 8 again, then restore with the commands above.

When something goes wrong

What you see	What to do
The browser says the connection is not secure	The domain did not point to the server at the first start. Check step 7 with `dig`, then on the server: `$C restart caddy`. The certificate arrives within minutes.
UptimeRobot says the site is down	Enter the server and run `$C ps`. If a part is not "Up": `$C up -d`. Read `$C logs --tail 100 api` if it keeps stopping.
The deploy ends with "The API did not start"	The last lines of the log are printed. Most often a typo in the settings: `nano /opt/ngclass/infra/.env`, fix, then run the deploy again.
`rsync: command not found` during the deploy	On the server: `sudo apt install -y rsync`, then deploy again.
`Permission denied (publickey)`	The key is not on the server: redo the box of step 6 with `ssh-copy-id`. If you lost the Mac key, use **KVM** in the OVH panel to log in and add a new one.
The site is slow	Check `free -h` and `df -h /`. With 100 users this should not happen; if the disk is over 80 % full, delete old backups or move to VPS-2.
A new version does not show	Reload the page once (Cmd + Shift + R on a computer). The entry files are never cached, so it should come by itself at the next visit.

In this table, `$C` means `sudo docker compose -f docker-compose.prod.yml`, run in `/opt/ngclass/infra`.

After the test month

The client…	What you do
continues	Nothing to change. OVH charges the server each month and the domain each year.
stops	Copy the last backups to the Mac (see Backups). Then in the OVH panel, on the VPS: **Résilier** (cancel), and on the domain turn off the automatic renewal if you do not need it.

PART E

Safety rules

- **Never share** the OVH password, the owner password, the teacher's temporary password (except with her, privately) or the file `~/.ssh/ngclass_ovh`. Only the IP and the domain are public.
- **The settings file** `/opt/ngclass/infra/.env` stays on the server. It holds the database password and the security secret. Never copy it into a message, a chat or GitHub.
- **Back up the key**: copy `~/.ssh/ngclass_ovh` to a USB stick kept at home. Without it you need the OVH KVM console to get back in.
- **Turn on two-step login** on the OVH account: it controls the server, the domain and the invoices.
- **Keep the domain renewal on** if the client continues. An expired domain takes the platform offline even if the server works.
- **Copy the backups to the Mac once a month** (see Part D): a third copy, outside OVH.
- **Watch the OVH emails**: invoices, payment failures and maintenance notices arrive there.

What was tested before you buy

On 12 November 2026 the exact production setup was started on the Mac and checked: HTTPS, the health check, the home page and deep links, the www redirect, the owner account and its first-login password change, a teacher created by the owner, a subject, a real file upload and download, the nightly backup, a full rebuild with the data still there, and a database restore.

> **Summary**
>
> Buy: SSH key (1), OVH account (2), VPS-1 monthly (3), .com domain (4). Configure: test SSH (6), domain to IP (7), `./infra/deploy.sh ubuntu@IP` (8), sign in (9), UptimeRobot (10), OVH backup (11). Updates: the same deploy command.